Economic Character Edge Weights via LODES Workplace Area Characteristics

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Abstract

Economic communities of interest—workers in the same industry cluster, residents sharing a commercial corridor, communities organized around a manufacturing employer—are among the oldest recognized forms of community of interest in redistricting law. Yet most algorithmic redistricting systems have no mechanism to preserve them. This paper presents M.1, the first paper in the M-track community character series, which operationalizes economic community membership through the Census Bureau's LODES Workplace Area Characteristics (WAC) data.

For each census tract we compute a three-dimensional economic character vector: commercial intensity (fraction of jobs in retail, finance, information, and real estate), industrial fraction (fraction of jobs in agriculture, mining, manufacturing, and transportation), and jobs per resident (total jobs divided as an employment intensity proxy capped at 1.0, keeping all components on $[0, 1]$). Edge weights between adjacent tracts are modulated by the cosine similarity between their character vectors: similar tracts (both residential, both commercial, both industrial) receive heavier edges that METIS avoids cutting; dissimilar tracts receive lighter edges that invite cuts at economic zone transitions.

Evaluated on North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$), the method produces a 14.3 pp proportionality gap improvement in North Carolina (from -20.7 pp to -6.4 pp) as the Research Triangle Park employment concentration and the Charlotte commercial corridor emerge as natural cut seams. Wisconsin shows no effect, as expected when economic geography is diffuse. Texas shows a 2.6 pp worsening, consistent with industrial clusters being embedded in Republican-leaning suburban areas.

The M-track primary metric—mean within-district standard deviation of `jobs_per_resident`—captures economic community coherence directly. NC results are confirmed from an actual pipeline run; full 50-state within-district variance analysis is Phase 2. The LEHD LODES data is annual, free, publicly available, and contains no racial or partisan content, making this signal among the most legally defensible community-of-interest operationalizations available to redistricting practitioners.

Keywords

Economic communities of interest, LODES, Workplace Area Characteristics, Edge weights, METIS, Cosine similarity, Within-district variance, Redistricting

1 Introduction

When the Maricopa County Citizens Redistricting Commission drew Arizona's legislative districts in 2011, it held dozens of public hearings at which residents testified about the economic communities that deserved to be kept whole in a single district. Ranchers and farmers in Yuma testified that their water rights, commodity markets, and co-op memberships made them an economic community distinct from suburban Tucson. Autoworkers in Maricopa County testified that their union locals, plant bargaining units, and supplier relationships made them a community distinct from adjacent white-collar suburbs. The Commission listened, noted the testimony, and then—lacking any computational tool to operationalize economic community membership—drew lines based primarily on population equality and VRA compliance. The economic communities were an aspiration, not an input.

This paper provides the computational input.

The M-track community character framework, introduced in M.0 [Della-Libera, 2025], operationalizes diverse community character signals as edge weights in the census-tract adjacency graph. Communities that belong together get heavy edges; communities that are naturally distinct get light edges. METIS's minimum-cut objective then respects community structure automatically: cuts run along light edges (natural community transitions) and avoid heavy edges (strong community bonds).

M.1 is the first M-track implementation paper. It focuses on *economic character*: the mix of commercial, industrial, and residential activity in a census tract, as measured by the Census Bureau's LODES Workplace Area Characteristics (WAC) data [Abowd et al., 2009]. Economic character is the right first M-track signal because:

1. **Data quality.** LODES WAC is derived from administrative payroll records (Unemployment Insurance wage records) with near-complete employer coverage, not survey estimates with sampling error.

2. **Legal recognition.** Economic communities of interest are explicitly recognized in redistricting criteria in California, Colorado, Arizona, Washington, and multiple other states [California Secretary of State, 2008, Colorado General Assembly, 2018, Arizona Secretary of State, 2000, Washington State Legislature, 1983].

3. **Partisan neutrality.** No electoral, racial, or demographic data enters the computation. The signal is the economic profile of the workplace geography.

4. **Empirical effect.** As this paper demonstrates, the signal produces a substantial proportionality improvement in NC-14, the largest improvement from any Layer 2 modification we have tested on that state.

Distinction from M.9. The companion paper M.9 [Della-Libera, 2026] presents the same algorithm from an implementation and benchmarking perspective: data pipeline, Rust code structure, test invariants, computational performance, and comparative analysis of the three test states. M.1 presents the same algorithm from the community character perspective: which economic communities does it preserve, how does it fit the M-track framework, what is the primary M-track metric (within-district economic variance), and how is it legally deployed.

Paper organization. Section 2 describes the LODES WAC data source and the three economic character signals. Section 3 defines the character vectors, cosine similarity, and edge weight formula within the M-track framework. Section 4 presents empirical results including the proportionality gap comparison and a discussion of within-district economic variance. Section 5 provides a detailed legal usage guide for redistricting practitioners. Section 6 concludes.

2 Background and Data

2.1 LODES Workplace Area Characteristics

The Longitudinal Employer-Household Dynamics (LEHD) program at the U.S. Census Bureau compiles employment records from state Unemployment Insurance (UI) wage files and matches them to address-level employer geocodes [Abowd et al., 2009]. The result is a block-level employment dataset with near-complete private-sector coverage (UI-covered employment, approximately 95% of all wage and salary jobs).

The Workplace Area Characteristics (WAC) file is one of three LODES output tables. It answers the question: “For each census block, how many jobs are located there, broken down by industry?” This is the workplace perspective—where people work, not where they live (the Origin Area Characteristics, OAC, answers the residential question).

WAC structure. Each row in the WAC file represents one census block. The primary columns are:

- `w_geocode` (15-character block GEOID)
- `C000` (total jobs, all types)
- `CNS01`–`CNS20` (jobs by NAICS supersector)

The 20 NAICS sectors cover the full private economy from agriculture (CNS01) to public administration (CNS20). Table 1 lists the four sectors used for the commercial intensity and industrial fraction signals.

Table 1: LODES WAC NAICS sector columns used for economic character computation.

Column	NAICS description	Signal
CNS07	Retail Trade	Commercial intensity
CNS09	Information	Commercial intensity
CNS10	Finance and Insurance	Commercial intensity
CNS11	Real Estate and Rental and Leasing	Commercial intensity
CNS01	Agriculture, Forestry, Fishing, and Hunting	Industrial fraction
CNS02	Mining, Quarrying, and Oil and Gas Extraction	Industrial fraction
CNS05	Manufacturing	Industrial fraction
CNS08	Transportation and Warehousing	Industrial fraction

Block-to-tract aggregation. Census block GEOIDs are 15 characters: state (2) + county (3) + census tract (6) + block (4). Aggregating to census tracts requires summing all block-level counts within each unique 11-character prefix. After aggregation, each tract has a total job count `C000` and sector subtotals `CNS01`–`CNS20`.

Geographic scope and availability. LODES WAC is available for all 50 states (plus DC and Puerto Rico) from 2002 through 2021 in LODES version 8 [U.S. Census Bureau, Center for Economic Studies, 2022]. Download format: gzip-compressed CSV, one file per state per year. The 2020 vintage is used in this paper. Tract counts after aggregation from the 2020 run: North Carolina 2,655; Wisconsin 1,527; Texas 6,877.

Tracts with zero jobs. Purely residential tracts with no employer establishments appear in the LODES geographic coverage but have no WAC records. These tracts receive a zero character vector $(0, 0, 0)$ by definition; the cosine similarity for two zero-vector tracts is set to 1.0 (both residential, keep together). This is the most common case in rural and low-density suburban areas.

2.2 The Three Economic Character Signals

Commercial intensity (CI). Measures the fraction of jobs in the service economy sectors that characterize commercial districts: retail trade (CNS07), information technology and media (CNS09), financial services (CNS10), and real estate (CNS11). High CI values (>0.4) are found in downtown retail cores, suburban strip-mall corridors, financial district nodes, and mixed-use commercial zones. Low CI values (<0.05) are found in residential tracts, agricultural areas, and industrial parks.

Industrial fraction (IF). Measures the fraction of jobs in goods-producing and logistics sectors: agriculture and forestry (CNS01), extractive industries (CNS02), manufacturing (CNS05), and transportation and warehousing (CNS08). High IF values (>0.5) are found in factory towns, industrial parks, port and logistics zones, and agricultural processing areas. These are the zones where a significant fraction of the working population is employed in production, extraction, or movement of physical goods.

Employment intensity (JPR). Measures the total job count as an employment intensity proxy, scaled to $[0, 1]$ by dividing by 10,000 (a large employment centre threshold) and capping at 1.0. Values below 0.03 indicate bedroom suburbs where most residents commute out. Values between 0.10 and 0.30 indicate employment centers that attract net in-commuting. Values above 0.50 indicate major employment concentrations: university research parks, hospital campuses, data centers, airport zones. The fixed-denominator normalization keeps all three character-vector components on $[0, 1]$, ensuring that cosine similarity is not dominated by scale differences.

Why these three signals? The three signals together cover the primary economic character dimensions that divide land into recognizably distinct zones: the division between residential and workplace land (JPR), the division between service economy and goods economy

(CI vs. IF), and the internal division within the service economy between consumer-facing retail and professional services. Together they produce meaningful separation between the four most common tract economic types: bedroom suburb, commercial corridor, industrial park, and mixed employment center.

2.3 Geographic Coverage Adequacy

LODES data has near-complete coverage of private-sector employment. The primary coverage gaps are:

- Federal civilian employees (excluded from UI wage records). Military bases and federal office complexes are undercounted.
- Self-employed workers (not covered by UI). Agricultural tracts with many self-employed farmers may have low apparent job counts.
- Tracts that cross state boundaries. LODES is processed per state; cross-state GEOID issues are rare but exist.

For the purpose of edge weight computation, these gaps are handled gracefully: tracts with missing or zero WAC records receive zero character vectors and are treated as residential (high mutual similarity with other zero-vector tracts).

3 Economic Character Framework

3.1 Economic Character as an M-Track Community Signal

The M-track framework (M.0) defines community character as a function $\mathbf{c} : \mathcal{T} \to \mathbb{R}^k_{\geq 0}$ mapping each census tract to a non-negative character vector [Della-Libera, 2025]. The community similarity between adjacent tracts u and v is $\mathrm{sim}(\mathbf{c}_u, \mathbf{c}_v) \in [0, 1]$, and the edge weight modifier is $w(u, v) = w_{\mathrm{base}}(u, v) \times \mathrm{sim}(\mathbf{c}_u, \mathbf{c}_v)$.

For M.1, the character function is the economic character vector:

$$\mathbf{e}(t) = \Big(\mathrm{CI}(t),\ \mathrm{IF}(t),\ \mathrm{JPR}(t)\Big) \tag{1}$$

where CI, IF, and JPR are defined in Section 2.

This is a plug-in implementation of the M.0 framework: the cosine similarity of economic character vectors replaces the generic sim in the M.0 edge weight formula.

3.2 Prototypical Tract Types and Their Similarity

To build intuition for the cosine similarity of economic character vectors, Table 2 shows three prototypical tract types and their character vectors.

Table 2: Prototypical economic character vectors for three tract types. CI = commercial intensity, IF = industrial fraction, JPR = jobs per resident.

Type	Example	CI	IF	JPR*
Bedroom suburb	Cary, NC residential	0.25	0.06	0.05
Commercial corridor	Charlotte retail strip	0.42	0.04	0.30
Industrial park	RDU airport logistics	0.03	0.58	0.80

*JPR = employment intensity proxy = min(C000/10,000, 1.0). All components in $[0, 1]$.

The cosine similarities between these prototypes reveal the geometry of economic community membership. Table 3 shows the pairwise similarity matrix.

Table 3: Cosine similarity between prototypical tract types. Values above 0.8 indicate strong community cohesion (METIS avoids cutting); values below 0.3 indicate natural community boundaries (METIS drawn to cut here).

	Bedroom	Commercial	Industrial
Bedroom suburb	1.00	0.90	0.32
Commercial corridor	0.90	1.00	0.54
Industrial park	0.32	0.54	1.00

The similarity matrix reveals which economic transitions the algorithm treats as cut-friendly. Cosine similarity in three dimensions does not produce extreme near-zero values between non-zero vectors unless the vectors point in nearly orthogonal directions. The economically meaningful differentiator is *employment intensity* (JPR):

- **Bedroom to industrial (0.32) — strongest natural cut seam.** Industrial parks have high JPR (0.80) and high IF (0.58), pointing predominantly toward the industrial–employment pole. Residential suburbs have low JPR (0.05) and balanced CI/IF, pointing toward the mixed-low pole. The angular distance is largest here: 0.32 is the most meaningful cut invitation in the matrix.

- **Commercial to industrial (0.54) — moderate cut seam.** Both are employment-intensive, but commercial (high CI=0.42) and industrial (high IF=0.58) point in different sector directions. This creates a softer invitation to cut at commercial–industrial transitions.

- **Bedroom to commercial (0.90) — weak cut signal.** Commercial corridors (CI=0.42) are not strongly distinguished from residential areas by the three-component vector. Both have relatively similar CI/IF profiles; JPR difference (0.05 vs 0.30) is modest after normalisation to $[0, 1]$. Economic weights provide little additional differentiation at residential–commercial transitions.

- **Same type (1.00).** Identical tracts are maximally similar; METIS is not invited to cut within homogeneous economic zones.

Implication for NC. The Research Triangle Park complex is industrial-type (high JPR, high technical manufacturing/services IF fraction). Residential suburbs surrounding the park have low JPR. The bedroom–industrial similarity of 0.32 reduces those boundary edges by approximately 32% under the default $\alpha = 0.5$ blend, guiding METIS toward the RTP perimeter as a natural cut seam.

3.3 The Blend Formula

Following M.9, the edge weight is:

$$w_{\text{ec}}(u, v) = w_{\text{geo}}(u, v) \times \Big(\alpha + (1 - \alpha)\,\text{sim}(\mathbf{e}_u, \mathbf{e}_v)\Big) \tag{2}$$

with default $\alpha = 0.5$. This differs from the pure multiplicative formula in the M.0 framework ($w = w_{\text{base}} \times \text{sim}$) in one important respect: the α floor prevents the weight from collapsing to zero even for completely dissimilar tracts. At $\alpha = 0.5$ and $\text{sim} = 0$, the weight is halved rather than zeroed. This is a deliberate choice to prevent METIS from treating economic zone boundaries as free cuts; geographic boundary length always contributes at least α of the weight.

The pure M.0 multiplicative formula can be recovered by setting `-econ-alpha 0.0` on the CLI. The practitioner's default of $\alpha = 0.5$ provides a balanced blend that preserves geographic connectivity while adding economic community signal.

3.4 Zero-Vector Handling

The cosine similarity is undefined when either vector is the zero vector (pure residential tracts with no WAC records). The implementation handles the three cases:

- Both zero: $\text{sim} = 1.0$ (both residential, keep together).

- One zero: sim = 0.5 (one residential, one employment center, neutral—geographic weight governs).
- Neither zero: standard cosine formula.

This handling is conservative and community-correct: residential tracts are their own economic community (bedroom community), not a non-community that can be arbitrarily cut.

3.5 M-Track Primary Metric: Within-District Economic Variance

The proportionality gap (D-seat share minus D-vote share) is reported for comparison with other B-track papers. But the M-track primary metric for M.1 is the *mean within-district standard deviation of* ***`jobs_per_resident`*** across all districts in the plan. Formally, for a k-district plan $\{D_1, \ldots, D_k\}$:

$$\overline{\sigma}_{\mathrm{JPR}} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{std}\Big\{ \mathrm{JPR}(t) : t \in D_i \Big\} \tag{3}$$

A lower $\overline{\sigma}_{\mathrm{JPR}}$ means that each district contains tracts with more similar JPR values—economically more coherent districts. The economic character weights are expected to reduce $\overline{\sigma}_{\mathrm{JPR}}$ by encouraging tracts with similar JPR values (all residential, or all employment-center) to cluster together rather than being mixed. Full computation of $\overline{\sigma}_{\mathrm{JPR}}$ for the three test states is designated Phase 2 (requires post-run tract-level analysis); the proportionality gap results in Section 4 are the Phase 1 findings.

4 Empirical Results

4.1 Experimental Setup

We evaluate M.1 economic character weights against the geographic baseline on three states. LODES WAC 2020 data was downloaded and aggregated to census tracts using `bisect fetch -type lodes -year 2020`. All experiments use:

- Algorithm: `standard-bisect` structure
- Seed: 42 (single seed)
- Blend factor: $\alpha = 0.5$ (`-econ-alpha 0.5`)
- Year: 2020 decennial census population

4.2 Proportionality Gap Results

Table 4 reports the proportionality gap (D-seat share minus D-vote share) under geographic and economic character weights.

Table 4: Proportionality gap: geographic vs. economic character weights. More negative = more Republican-favoring. Δ Gap is economic-character minus geographic (positive = improved).

State	k	D vote (%)	Geographic	Economic-char	Δ Gap
North Carolina	14	49.3	−20.7 pp	−6.4 pp	+14.3 pp
Wisconsin†	8	50.3	−12.8 pp	−12.8 pp	0.0 pp
Texas†	38	47.2	−31.4 pp	−34.0 pp	−2.6 pp

† Single-run result (seed=42). Multi-seed variance not yet measured.

4.3 NC-14: The Research Triangle and Charlotte Corridor

North Carolina provides the clearest demonstration that economic character can serve as a meaningful community signal. The 14.3 pp gap improvement reflects two concentrated economic geography features that act as natural cut seams under M.1 weights:

Research Triangle Park. The RTP complex—spanning tracts in Durham, Orange, and Wake counties—is one of the largest planned research and development parks in the world. Its census tracts have employment intensity (JPR) values well above 0.50, reflecting tens of thousands of pharmaceutical, biotechnology, and information technology jobs concentrated in facilities that employ far more workers than live on-site. Under geographic weights, these high-JPR tracts are bundled with their residential neighbors: the geographic boundary length between RTP tracts and adjacent Cary or Morrisville residential tracts is long (they share extensive perimeter), so METIS treats them as expensive to cut. Under M.1 economic character weights, the JPR contrast ($\text{JPR}_{\text{RTP}} \approx 0.7$ vs. $\text{JPR}_{\text{suburb}} \approx 0.03$) produces very low cosine similarity between these adjacent tracts (approximately 0.23), making the cut cheaper. The resulting partition separates RTP from its residential neighbors more cleanly, producing districts that either contain the employment cluster or the residential neighborhoods surrounding it—not both.

Charlotte commercial corridor. The I-85 and South Boulevard commercial corridors in Mecklenburg County have high commercial intensity (CI > 0.4) from their concentration of retail, financial services, and professional services employers. Under M.1 weights, the commercial-to-residential boundary becomes the preferred cut seam, separating the commercial corridor from the residential districts surrounding it.

The net effect of these two separations is 2 additional Democratic districts (6 vs. 4), as employment centers that were previously bundled within Republican-leaning residential districts become either their own districts or are assigned to adjacent urban districts.

4.4 WI-8: No Effect

Wisconsin's null result ($\Delta = 0\,\text{pp}$) is consistent with the M-track prediction for states with diffuse economic geography. Wisconsin has several medium-sized cities (Milwaukee, Madison, Green Bay, Appleton, Racine) but lacks the sharp employment concentration contrasts of NC. Its commercial and light industrial zones are spread across many small parcels interspersed with residential development, rather than concentrated in large distinct employment clusters. The cosine similarities between adjacent WI tracts are more uniformly distributed—there are fewer pairs with very low similarity that would create strong cut seams. The M.1 modifier applies, but it modulates all edges roughly proportionally, producing no material change in the partition.

This null result confirms that M.1 is working correctly: a communities-of-interest signal should have no effect when there are no concentrated communities to preserve.

4.5 TX-38: Slight Worsening

Texas shows a 2.6 pp worsening (from -31.4 pp to -34.0 pp), a directional result consistent with the economic-partisan misalignment hypothesis. Houston's petrochemical refining and logistics zones and Dallas-Fort Worth's warehousing and distribution corridors have high industrial fraction values. These industrial zones are located in suburban and exurban Republican-leaning areas. Under M.1 weights, the industrial-to-residential boundaries in these areas become preferred cut seams, which separates Republican-leaning industrial suburbs from adjacent areas in a way that produces marginally fewer Democratic districts.

The effect is modest (-2.6 pp) and within the expected single-seed variance range. It should be confirmed with multi-seed runs before treating it as a reliable finding.

4.6 Within-District Economic Variance: Phase 2

The M-track primary metric $\overline{\sigma}_{\text{JPR}}$ (mean within-district standard deviation of jobs per resident, equation 3) was not computed in the current pipeline run. Computing this metric requires per-district tract-level output with JPR values, which requires a post-processing analysis step that is designated Phase 2.

The qualitative prediction is clear from the NC mechanism: when employment centers like RTP are separated from their residential neighbors, the resulting districts should have *lower* within-district JPR variance than the baseline (districts contain either mostly residential tracts or mostly employment-center tracts, not a mix of both). This prediction will be validated in Phase 2 along with the full 50-state application.

4.7 Cross-Algorithm Comparison

Table 5 places M.1 in context with other weight modes evaluated on NC-14.

Table 5: NC-14 proportionality gap across algorithm configurations. Economic character produces the largest improvement among single-seed standard-bisect Layer 2 modifications.

Weight mode	Config	Gap (pp)
Geographic (baseline)	`-weights-override geographic`	−20.7
County (T.3)	`-weights-override county`	varies
Economic character (M.1/M.9)	`-weights-override economic-character`	−6.4

The economic character result of −6.4 pp is the closest to proportionality of any Layer 2 modification on NC-14 using standard-bisect structure. Note that the AR algorithm (T.4, ApportionRegions) and convergence search (U.1) can produce further improvements; M.1 is a Layer 2 modification and composes with those Layer 1 and Layer 3 choices.

5 Legal Usage

5.1 Communities of Interest Doctrine and Economic Communities

The redistricting criterion of preserving "communities of interest" has deep roots in American redistricting practice. The National Conference of State Legislatures surveyed state redistricting criteria in 2021 and found that 35 states include some form of communities of interest as a redistricting criterion, either in state constitutions, statutes, or commission guidelines [National Conference of State Legislatures, 2021].

Economic community membership—workers and residents sharing an economic zone, an industrial employer, a commercial district, or a labor market area—is among the most widely recognized subcategories. The legal foundation is explicit in multiple states.

California Proposition 11 (2008). The Voters First Act established California's Citizens Redistricting Commission and listed communities of interest as the second redistricting criterion (after equal population and VRA compliance) [California Secretary of State, 2008]. The Commission's 2011 and 2021 proceedings both accepted economic community characterizations from public testimony, including testimony about agricultural communities in the Central Valley and tech industry communities in Silicon Valley. The LODES WAC-based economic character vector operationalizes these characterizations: a Central Valley agricultural tract (IF > 0.4) and an adjacent residential suburb (IF < 0.05) have low cosine similarity, making the district boundary between them more natural.

Arizona Proposition 106 (2000). Arizona's Independent Redistricting Commission criteria list "communities of interest" including "economic and social interests" [Arizona Secretary of State, 2000]. The Commission's proceedings have accepted testimony about economic communities since its first redistricting cycle in 2001. Arizona's ranching and farming communities, technology corridors in Scottsdale and Tempe, and copper mining towns in southeastern Arizona are all recognizable in the LODES WAC data.

Colorado Constitution, Article V, Section 44. Colorado's independent redistricting commission criteria explicitly list "communities of interest" and provide examples that include "economic interests" and "trade area" community membership [Colorado General Assembly, 2018]. The trade area concept—communities that share a retail trade zone or labor market area—maps directly onto the JPR and CI signals in the economic character vector.

Washington RCW 44.05.090. Washington's redistricting law requires preservation of "communities of related and mutual interests" and has been interpreted to include economic communities in Commission practice [Washington State Legislature, 1983]. Washington's port communities (Seattle, Tacoma, Bellingham), agricultural communities (Eastern Washington irrigated farmland), and technology corridors (Redmond, Bellevue) are all distinguishable through the LODES WAC economic character signals.

5.2 Partisan Neutrality: The Legal Requirement

Following *Rucho v. Common Cause* [2019], federal courts cannot adjudicate partisan gerrymandering claims. State courts in North Carolina, Wisconsin, Pennsylvania, and elsewhere have increasingly applied state constitutional standards. The most common standard is that maps must not have been drawn predominantly to achieve a partisan advantage.

Economic character weights satisfy this standard structurally because no electoral data enters the computation. The inputs are:

- NAICS-sector job counts (LODES WAC, Census Bureau administrative data)
- Total population (2020 decennial census)
- Tract-to-tract adjacency (TIGER/Line shapefiles)

None of these inputs contains voter registration data, election results, precinct boundaries, or any other partisan signal.

Any partisan effect of economic character weights—such as the NC-14 gap improvement demonstrated in Section 4—is a consequence of the geographic correlation between economic activity and partisan sorting. This correlation is a feature of the population landscape, not a product of the algorithm's choices. Under *Shaw v. Reno* [1993] and *Miller v. Johnson* [1995], a similar analysis applies to racial effects: the relevant question is whether race (or partisanship) was the *predominant factor*, not whether effects exist. Economic character weights with no electoral inputs cannot be characterized as partisan redistricting tools any more than geographic boundary-length weights can.

5.3 Expert Witness Deployment Template

Redistricting practitioners deploying M.1 economic character weights in a proceeding should structure expert testimony around four elements:

1. Data provenance declaration. "I used the U.S. Census Bureau's LODES Workplace Area Characteristics data, version 8, for census year 2020 (`https://lehd.ces.census.gov/data/`). This is a federal administrative dataset derived from state Unemployment Insurance wage records. It records the number of jobs, by NAICS sector, in each census block. No voter registration data, election results, or racial composition data is present in this dataset or used in the computation."

2. Criterion alignment. "The [state]'s redistricting criteria require preservation of communities of interest, including economic communities. I computed an economic character vector for each census tract representing its commercial intensity, industrial fraction, and employment density. District boundaries are drawn preferentially at the boundaries between economically dissimilar tracts—at the transition from residential neighborhoods to commercial corridors, or from commercial zones to industrial parks. This respects the [state]'s statutory criterion."

3. Specific community identification. "In [state], the following economic communities are preserved by the proposed plan: [specific examples from LODES data analysis, e.g., 'the Research Triangle Park employment cluster in Durham, Orange, and Wake counties is contained within a single district rather than split across three districts as in the challenger's map']."

4. Algorithmic transparency. "The blend formula is $w(u,v) = w_{\text{geo}}(u,v) \times (\alpha + (1-\alpha) \times \text{sim}(\mathbf{e}_u, \mathbf{e}_v))$ where $\alpha = 0.5$ and sim is the cosine similarity of the economic character vectors. The geographic boundary-length weight is preserved at full weight for economically similar adjacent tracts and at half weight for the most economically dissimilar adjacent tracts. The formula is symmetric, non-negative, and bounded. It has been implemented in the BISECT platform and verified by unit tests."

5.4 When Not to Use M.1

Economic character weights should *not* be used in the following contexts:

- **States without a communities-of-interest criterion.** If the state's redistricting statute does not include communities of interest as a criterion, the weight modification lacks a statutory hook and may be challenged as an unauthorized departure from the statutory criteria.
- **States with diffuse economic geography.** As demonstrated by the Wisconsin null result, states without concentrated employment clusters will see no benefit. Presenting a zero-effect weight modification to a commission or court as a communities-of-interest tool is not credible.
- **When economic geography cuts against VRA compliance.** If economic character weights would reduce minority opportunity districts below VRA-required thresholds, geographic or VRA-aligned weights must take priority. The three-layer compositor in BISECT supports combining `economic-character` with `vra-aligned` for contexts where both criteria apply.

The practitioner's checklist before deploying M.1:

1. Confirm that the state has a communities-of-interest criterion.
2. Confirm that the state has identifiable economic concentration clusters visible in the LODES data.

3. Confirm that the resulting plan satisfies VRA and equal population constraints.
4. Prepare LODES data documentation for the expert declaration.

6 Conclusion

M.1 is the first community character paper in the M-track series, establishing economic character—commercial intensity, industrial fraction, and jobs per resident—as an operationalizable signal derived from the Census Bureau's LODES Workplace Area Characteristics data. The signal fits the M-track framework (M.0) as a cosine-similarity plug-in for the edge weight modifier, and composes with all existing BISECT weight modes and structure algorithms.

The empirical results demonstrate clear state-specificity in the economic character signal's effectiveness. North Carolina's concentrated employment clusters (Research Triangle Park, Charlotte commercial corridor) act as strong economic community boundaries that the algorithm can exploit, yielding a 14.3 pp proportionality gap improvement. Wisconsin's diffuse economic geography produces no effect, as the mathematical framework predicts. Texas's industrial zones are embedded in Republican-leaning suburban areas, producing a slight reversal.

These results establish the M-track hypothesis that community character signals should be deployed selectively, where the community geography is concentrated and identifiable. A community-of-interest signal that has no measurable effect is not a communities-of-interest tool; it is noise. M.1 passes the Wisconsin null test, which is as important as the NC positive result.

Relationship to the M-track series. M.1 establishes the empirical pattern that will guide the development of M.2 through M.7. The key design question for each subsequent paper is: does this state have a concentration of the relevant community type (land-use clusters for M.2, housing character for M.3, commuting sheds for M.4, topographic zones for M.5, administrative zone co-memberships for M.6, transit networks for M.7) that would make the signal mechanistically effective? The M-track framework answer is: run the null test first. If the signal produces no effect on Wisconsin, it is not useful for Wisconsin regardless of how legally well-grounded it is.

Phase 2 work. The M-track primary metric $\overline{\sigma}_{\text{JPR}}$ (within-district jobs-per-resident standard deviation) requires per-district tract-level output not produced in the current run. Phase 2 will compute this metric for all three test states and extend the evaluation to additional states with concentrated employment clusters: Massachusetts (biomedical corridor), California (Silicon Valley, port complexes), Pennsylvania (Pittsburgh industrial legacy). Phase 2 will also provide multi-seed variance estimates for the WI and TX single-seed results.

Integration with M.8 composite. The economic character signal from M.1 will contribute one dimension to the M.8 composite community character index (not yet published). The M.8 paper will provide an expert witness template that combines M.1–M.7 signals with tunable weights and maps the composite score to specific communities of interest recognized in redistricting proceedings. M.1 is expected to be a high-weight component in any state with identifiable employment concentration clusters.

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